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MTA4DPR: Multi-Teaching-Assistants Based Iterative Knowledge Distillation for Dense Passage Retrieval

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Abstract

[ILLEGIBLE]

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[ILLEGIBLE]

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Ethics Statement

[ILLEGIBLE]

Licenses

[ILLEGIBLE]

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A Algorithm 1

Algorithm 1 MTA4DPR Training Process

Require: T : the teacher model; TA : the assistant models; M_θ : the student model;
 Q : the query set; P : the passage set; max_iter : maximum number of training
iterations; max_steps : maximum number of training steps; η : Learning rate;

Ensure: M_θ

```
1:  $i \leftarrow 0$ 
2: while  $i < max\_iter$  do
3:    $D_{train}, D_{eval} \leftarrow GenDataset(T, TA, Q, P)$ 
4:   repeat
5:      $id_{bestTA} \leftarrow TASelect(D_{train})$ 
6:      $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{total}(D_{train}, M_\theta, id_{bestTA})$ 
7:   until  $max\_steps$  reached
8:    $outperformed\_TA \leftarrow Compare(M_\theta, TA, D_{eval})$ 
9:   if  $outperformed\_TA$  then
10:    remove  $Worst(TA)$ 
11:    add  $M_\theta$  into  $TA$ 
12:   end if
13:    $i \leftarrow i + 1$ 
14: end while
```
